

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Lab Sheet No: 10

Date: 15.06.2026

Title: IoT Sensor Data Logging System using ESP32 and Google Sheets.

Aims:

To design and implement an IoT-based data logging system using an ESP32 microcontroller and a temperature sensor, where real-time sensor data is transmitted over Wi-Fi and stored in a Google Sheets spreadsheet. This system demonstrates cloud-based data storage, real-time monitoring, and basic IoT integration using HTTP requests and Google Apps Script.

Task 01: IoT Sensor Data Logging System Setup

1. Project Setup in Wokwi / VS Code
 - a. Name the project **SensorData_ESP32_GoogleSheet**.
2. Circuit Design
 - a. Sensor Connection
 - i. Connect the DS18B20 (One-Wire Digital Temperature Sensor) to the ESP32.
3. Library Integration
 - a. WiFi (Built-in ESP32 library)
 - b. HTTPClient (Built-in ESP32 library)
 - c. OneWire (External library)
 - d. DallasTemperature (External library)

4. Google Sheets Configuration

- a. Create a new Google Sheet
- b. Name the sheet **ESP32_Sensor_Data**
- c. Create the following column headers:
 - i. Date
 - ii. Time
 - iii. Sensor Value
- d. Open Google Apps Script from the Extensions menu
- e. Write a script to receive HTTP GET/POST requests from ESP32
- f. Deploy the script as a Web App
- g. Set access permission to Anyone
- h. Copy the generated Web App URL

5. Programming Logic Development in Wokwi / VS Code

- a. Initialize Serial Monitor
- b. Initialize Wi-Fi connection
- c. Initialize the DS18B20 sensor
- d. Read temperature values at regular intervals
- e. Send sensor data to Google Sheets using an HTTP requestData should be sent automatically at a fixed time interval (e.g., every 2 seconds)
 - i. Each sensor reading should be stored as a new row in Google Sheets

6. Simulation & Testing in Wokwi / VS Code

- a. Simulate the circuit in Wokwi
- b. Verify temperature readings in the Serial Monitor
- c. Confirm real-time data entries in Google Sheets